

NISHANKAR SATHIYAMOCHAN

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🎓 [Google Scholar](#)

👤 Personal Statement

I am a computer vision researcher with experience in medical image analysis, explainable computer vision, and evaluation of image generation. My work includes attribution methods for vision transformers and CNNs, radiology foundation-model features, and efficient visual learning. For a PhD, I aim to study reliable 3D and multimodal perception, particularly spatial understanding under incomplete or uncertain observations.

💡 Research Interests

Computer Vision, Explainable AI (XAI), Trustworthy AI, Spatial Intelligence.

🎓 Education

University of Peradeniya

B.Sc (Hons) in Computer Engineering, CGPA: 3.50/4.00

Kandy, Sri Lanka
Nov 2018 – Dec 2023

Hartley College

GCE Advanced Level - Physical Sciences

Point Pedro, Sri Lanka
Jan 2009 – Aug 2017

- Results: Combined Mathematics (A), Chemistry (A), Physics (B)

📖 Publications

Journals

- J1.** S. Nishankar, N. Pathirana, P. Sanjeevani, A. G. Perera, and S. Thuseethan, “Does Explainability Transfer? A Controlled Benchmark of Attribution Methods on Vision Transformers and CNNs”, **Neuro-computing**, Elsevier | Under Review | [arXiv:2608.02396](#) Jul 2026
- J2.** S. Nishankar, P. Sanjeevani, A. Perera, and S. Thuseethan, “HiLRP: Conservation-Valid Attribution via Attention Primitives”, **Pattern Recognition**, Elsevier | Submitted | [arXiv:2609.01282](#) Aug 2026
- J3.** S. Nishankar, P. Sanjeevani, and A. Perera., “M3-Score: Fidelity, Memorization and Coverage as Separate Axes for Evaluating Generative Radiology Image Models”, **Biomedical Signal Processing and Control**, Elsevier | Submitted | [arXiv:2609.33769](#) Sep 2026
- J4.** T. Shyamalee*, S. Nishankar*, S. Thuseethan, C. Wimalasooriya, and Y. Sebastian, “Deep Semi-supervised Learning for Medical Image Analysis: A Survey” (*Equal Contribution), **NeuroComputing** | Accepted with minor edit Sept 2025
- J5.** Tharindu Godage, Sathiyamohan Nishankar, Selvarajah Thuseethan, Shanmuganathan Vasantapriyan, Zhongwei Liang, “Trends, Challenges and Future Directions in Deep Learning for Citrus Leaf Disease Recognition: A Survey”, **Artificial Intelligence Review**, Springer | Q1 | [DOI](#) Jun 2026
- J6.** S. Nishankar, V. Pavindran, T. Mithuran, S. Nimishan, S. Thuseethan, and Y. Sebastian, “ViT-RoT: Vision Transformer-Based Robust Framework for Tomato Leaf Disease Recognition”, **AgriEngineering**, MDPI, vol. 7, no. 6, p. 185 | Q1 | [DOI](#) Jun 2025
- J7.** S. Nishankar, T. Mithuran, S. Nimishan, V. Pavindran, and S. Thuseethan, “TOM-SSL: Tomato Disease Recognition using Pseudo-labelling based Semi-supervised Learning”, **AgriEngineering**, MDPI, vol. 7, no. 8, p. 248 | Q1 | [DOI](#) Jul 2025
- J8.** George, R., S. Nishankar, Thuseethan, S., Wimalasooriya, C., Sebastian, Y., Ragel, R. G., & Liang, Z., “U-FedTomAtt: Ultra-lightweight Federated Learning with Attention for Tomato Disease Recognition”, **Information Sciences** | Q1 | Under Review | [arXiv:2602.16749](#) Feb 2026
- J9.** George, R., S. Nishankar, Thuseethan, S., & Ragel, R. G., “UtVAA: Ultra-tiny Vision Transformer with Affix Attention for Mobile Image Classification”, arXiv preprint | Under Review | [arXiv:2606.14735](#) Jun 2026

- J10.** Balenthira Thanusanth, Selvarajah Thuseethan, **Sathiyamohan Nishankar**, Romiyal George, Roshan G. Ragel, Bimali S. Weerakoon, Ayesh Jayasinghe, and Ananthamoorthy Krishnamoorthy, “Architecture-aware Robustness Evaluation of Explainable Deep Learning for Breast Cancer Diagnosis”, **IEEE Transactions on Radiation and Plasma Medical Sciences** | Q1 | Under Review Jun 2026
- J11.** Thilini Bakmeedeniya, Selvarajah Thuseethan, Roshan G. Ragel, Palanisamy Vigneshwaran, and **Sathiyamohan Nishankar**, “From Unimodal to Multimodal Deep Physiological Signal Analysis in Healthcare: A Survey”, **Computer Science Review** | Q1 | Minor Revision Mar 2026
- J12.** Romiyal George, **Sathiyamohan Nishankar**, Selvarajah Thuseethan, Kandiah Pakeerathan, Roshan G. Ragel, “SLIF-Brinjal: An In-Field Leaf Dataset for Disease Recognition in Precision Agriculture”, **Scientific Data**, Nature Portfolio | Q1 | Minor Revision Feb 2026

Conferences

- C1.** M. Shakishnavi, K.P.N. Jayasena, and **S. Nishankar**, “A Unified Zero Trust Orchestration Framework for Kubernetes: Design, Integration, and Functional Verification”, **APAN Conference** | Accepted 2026
- C2.** M. Shakishnavi, K.P.N. Jayasena, and **S. Nishankar**, “A Multi-Layer Zero Trust Security Framework for Kubernetes”, **MERCon** | [DOI](#) 2026

Current Researches

- S. Nishankar**, et al., “**SSA-HQNN: Spatial Superpixel Attention-Guided Hybrid Quantum Neural Network for Image Classification**”, (Drafting). 2026
- Proposed SSA-HQNN, a hybrid quantum-classical architecture that uses learned spatial superpixel attention (SSA) to directly condition a trainable quantum encoding for image classification.
 - Evaluated across seven benchmarks (including medical imaging modalities like PathMNIST and PneumoniaMNIST), achieving up to 93.9% and 90.3% accuracy respectively while matching or exceeding the AUC of classical baselines.
 - Designed a highly parameter-efficient backbone requiring as few as 60K trainable parameters, achieving a $\sim 100\times$ reduction in model size compared to ResNet-18.

Honors & Awards

- Best Researcher Award (Faculty Level)** Sabaragamuwa University (2025)
- 1st Place:** Pre Aces Hackathon, Inter-University Coding Competition 2022

Professional Experience

- Part-time Research Intern** March 2026 – Aug 2026
University of Southern Queensland, Australia *Supervisor: Dr. Asanka Perera*
- Developing **M3-Score**, a new evaluation metric for diffusion-based generative models in medical imaging, built on features from a frozen radiology-pretrained vision transformer.
 - Working on **3D Computer Vision** 3D reconstruction.
 - XAI Benchmark Study.
 - HiLRP Universal Explanation for Vision Transformers.
- Lecturer** Apr 2025 – Aug 2026
Sabaragamuwa University of Sri Lanka *Dept. of Software Engineering*
- Teaching undergraduate courses and supervising final year research projects.
 - University Business Linkage (UBL) Coordinator:** Facilitating collaboration between industry partners.
 - Assistant Research Coordinator:** Managing departmental research activities.
- Lecturer** Apr 2024 – Apr 2025
University of Jaffna *Dept. of Computer Engineering*
- Instructor** Jan 2024 – Apr 2024
University of Peradeniya *Dept. of Computer Engineering*

- Freelance Software Developer

Remote (Self-Employed)

- Developed DUCAS AI to automate daily reporting in construction using Azure, GPT-4o, and FastAPI.

2022 – Present
- Software Engineering Intern

Infinity Innovators Pvt

- Developed WAKA, a full-stack online vehicle marketplace for New Zealand using React.js and AWS.

Dec 2022 – Apr 2023

Colombo, Sri Lanka

 Teaching Experience

- Sabaragamuwa University of Sri Lanka – Lecturer

- Taught: Artificial Intelligence, Computer Communication Networks, High Performance Computing (HPC), Cloud Computing, Game Design and Development, and Parallel and Distributed Computing

2025 – 2026 Aug
- University of Jaffna – Lecturer on Contract

- Taught: Embedded Systems, Computing, Digital Image Processing, and Applied Algorithms

2024 – 2025
- University of Peradeniya – Instructor

- Led instruction for: Artificial Intelligence, Computer Communication Networks, and Advanced Database Management Systems

2024

 Projects

- Efficient Transfer Learning and XAI for Histopathology

- Utilized XAI methods (SHAP, LIME, Grad-CAM) to enhance model trust.

2023
- Road Sign Detection in Low Light Conditions

- Developed an AI system using YOLOv8 to detect signs in low-light conditions.

2023

 Academic Service & Memberships

- Publication Chair

ComURS Symposium (2026), ICARC International Conference (2026)
- Open Track Co-Chair

ICARC International Conference (2026)
- Registration Chair

ICARC International Conference (2026)
- Editor

ICARC (2026), ComURS (2026), ICSUSL (2025)
- Reviewer

ICARC (2026), ComURS (2026), Moratuwa Engineering Research Conference (MERCon) (2026)
- Member

Institute of Electrical and Electronics Engineers (IEEE)

 Technical Skills

Programming: Python, Node.js, Express.js, Django, Flask, FastAPI, Java, C++, SQL
AI/ML: PyTorch, TensorFlow, Keras, OpenCV, Scikit-learn, YOLOv8, Hugging Face, Nerfstudio
Cloud/DevOps: AWS (Serverless, EC2, S3), Azure, Docker, Git, React.js, Flutter

Collaborations

Dr. S. Thuseethan

- Lecturer in Information Technology, Charles Darwin University, Australia
- Email: Thuseethan.selvarajah@cdu.edu.au

Dr. Asanka Perera

- Lecturer (Mechatronic Engineering) in University of Southern Queensland, Australia
- Email: Asanka.Perera@unisq.edu.au

Dr. Pubudu Sanjeewani

- Associate Lecturer, RMIT University, Melbourne, Australia
- Email: pubudu.sanjeewani@rmit.edu.au